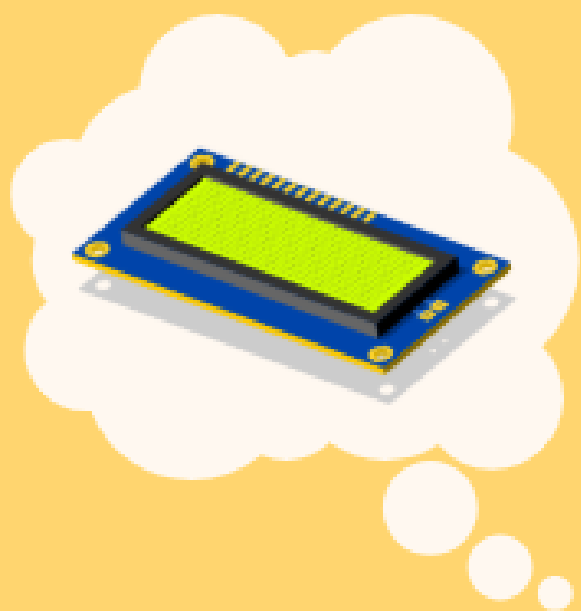


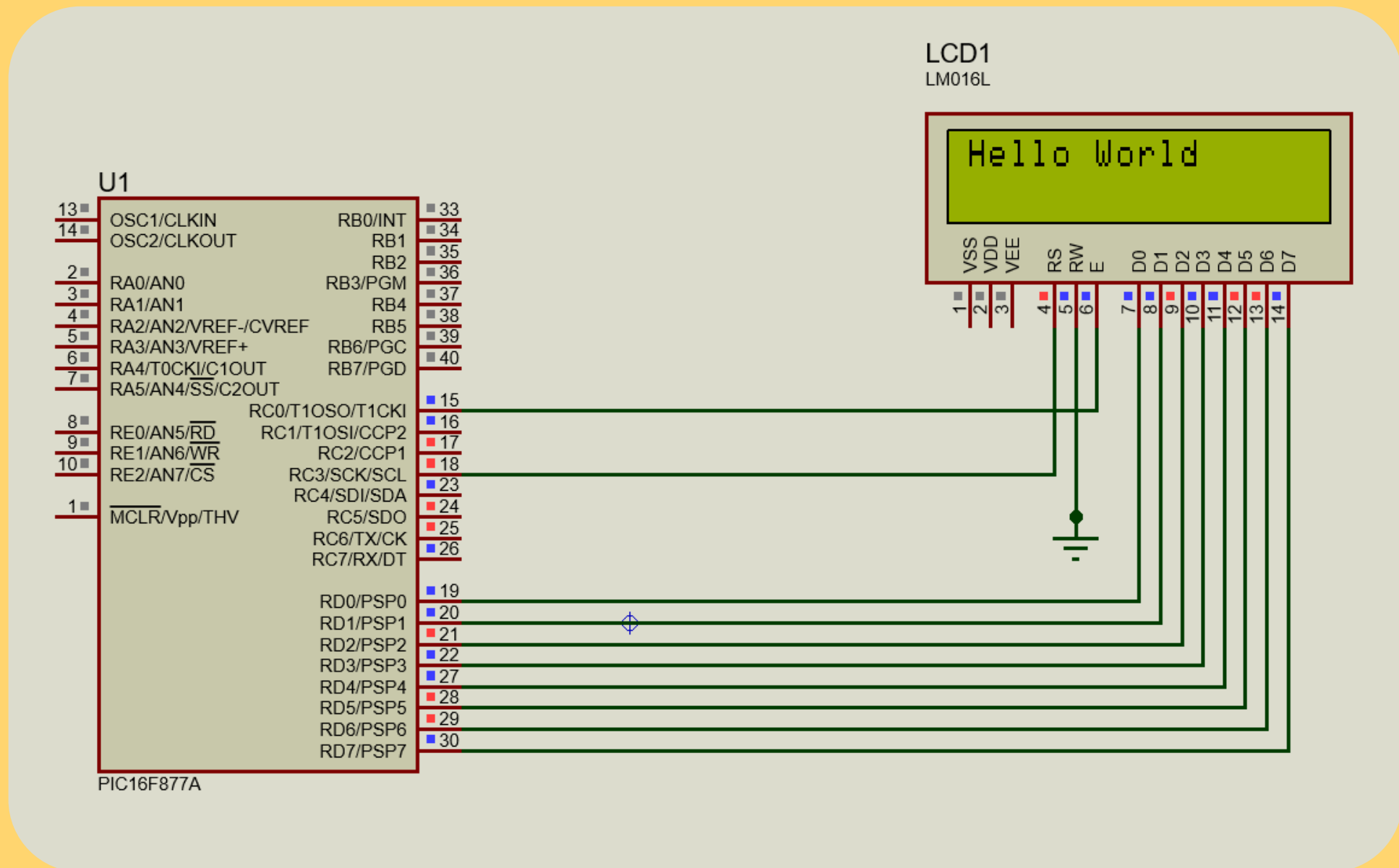


WORLD



Goal:

- To display “Hello World” on the LCD by sending data through PORTD.

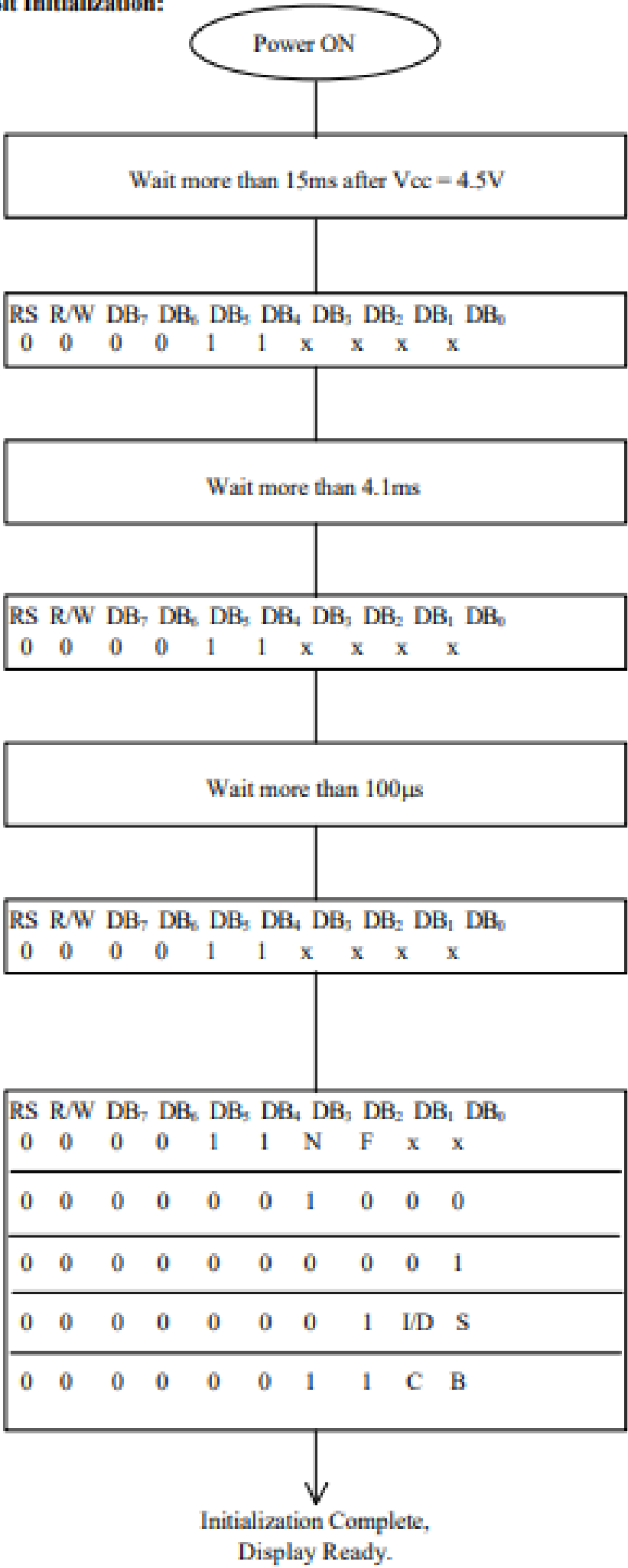


- For simplicity, we're sending 8-bit data through the LCD's data pins D0 to D7.
- We make the RW pin low (0) by grounding it to enable the write operation.

As the first step, we need to initialize the LCD.



8 - Bit Initialization:



No data should be transferred to or from the display during this time.

Function Set Command: (8-Bit interface)
BF cannot be checked before this command.

No data should be transferred to or from the display during this time.

Function Set Command: (8-Bit interface)
BF cannot be checked before this command.

No data should be transferred to or from the display during this time.

Function Set Command: (8-Bit interface)
After this command is written, BF can be checked.

Function Set (Interface = 8 bits, Set No. of lines and display font)

Display OFF

Clear Display

Entry Mode Set:

Display ON (Set C and B for cursor/Blink options.)

Note: BF should be checked before each of the instructions starting with Display OFF.



DB ₇	DB ₆	DB ₅	DB ₄	DB ₃	DB ₂	DB ₁	DB ₀
0	0	1	1	x	x	x	x

Function Set Command: (8-Bit interface)
BF cannot be checked before this command.

- Here, 'x' represents a don't-care condition. Assuming it as 0 gives a value of 0x30.

DB ₇	DB ₆	DB ₅	DB ₄	DB ₃	DB ₂	DB ₁	DB ₀
0	0	1	1	N	F	x	x

Function Set (Interface = 8 bits, Set No. of lines and display font)

Sets the interface data length, the number of lines, and character font.

	RS	R/W	DB ₇	DB ₆	...				DB ₁	DB ₀
Code	0	0	0	0	1	DL	N	F	x	x

Note: x = Don't Care

DL: Sets interface data length. Data is sent or received in 8-bit length (DB₇ ~ DB₀) when DL = "1", and in 4-bit length (DB₇ ~ DB₄) when DL = 0. When the 4-bit length is selected, data must be sent or received twice.

N: Sets the number of lines

N = "0" : 1 line display (1/8 duty)

N = "1" : 2 line display (1/16 duty)

F: Sets character font.

F = "1" : 5 x 10 dots

F = "0" : 5 x 7 dots

- For 8-bit data length, DL = 1; for 2-line display, N = 1; using 5×7 dots, F = 0; and considering x = 0.
- After setting all configuration bits, the final function set value becomes 0x38.

Clear Display

	RS	R/W	DB ₇	DB ₆	...	DB ₁	DB ₀	
Code	0	0	0	0	0	0	0	1

- The command 0x01 is used to clear the LCD display.



	RS	R/W	DB ₇	DB ₆	...	DB ₁	DB ₀
Code	0	0	0	0	0	1	D C B

Controls the display ON/OFF status, Cursor ON/OFF and Cursor Blink function.

D: The display is ON when D = 1 and OFF when D = 0. When OFF due to D = 0, display data remains in the DDRAM. It can be displayed immediately by setting D = 1.

C: The cursor displays when C = 1 and does not display when C = 0. The cursor is displayed on the 8th line when 5 x 7 dot character font has been selected.

B: The character indicated by the cursor blinks when B = 1. The blink is displayed by switching between all blank dots and display characters at 0.4 sec intervals.
The cursor and the blink can be set to display simultaneously.

- Display should be ON (D = 1). Since we're not using the cursor, C = 0; and as blinking is disabled, B = 0. i.e 0xC0

With all these configurations applied, the LCD is now successfully initialized.

```
void init(){
    TRISC = 0x00; //setting all pins as output
    TRISD = 0x00; //setting all pins as output
    _delay_ms(100);
    lcd_command(0x30); //Function Set Command for 8 bit
    _delay_ms(100);
    lcd_command(0x30); //Function Set Command for 8 bit
    _delay_ms(100);
    lcd_command(8); //Function Set Command for 8 bit
    lcd_command(0x38); //Set lines(N) & font(F)
    lcd_command(0x0C); //Display on, cursor off, blink off
    lcd_command(0x01); //Clear Display
}
```

The LCD is initialized. But what about **lcd_command** function? It's not a built-in function, we create it manually to send commands to the LCD.



Why don't these values (like 0x30) appear on the LCD?

How are they recognized as commands instead of display characters?!

- This is where the RS pin of the LCD comes into play — it decides whether the value is a command or data.
- When RS = 0, it's a command; when RS = 1, it's data.

```
void lcd_command(unsigned char i){  
    RC3 = 0;    //RS pin set high (command mode)  
    PORTD = i;  
    RC0 = 1;    //Enable pin for clock set high  
    delay_ms(100);  
    RC0 = 0;    //Enable pin for clock set low  
}
```

```
void lcd_data(unsigned char i){  
    RC3 = 1;    //RS pin set high (data mode)  
    PORTD = i;  
    RC0 = 1;    //Enable pin for clock set high  
    delay_ms(100);  
    RC0 = 0;    //Enable pin for clock set low  
}
```

- We use the E pin to latch data into the LCD by making it HIGH and then LOW, creating a short enable pulse.

```
#include <xc.h>  
#define _XTAL_FREQ 6000000  
void init();  
void lcd_command(unsigned char );  
void lcd_data(unsigned char);  
unsigned char array[18]="Hello World";  
void main(void) {  
    init(); //Initialization of LCD  
    lcd_command(0x80); // Setting initial location  
    for(int i=0;i<11;i++){  
        lcd_data(array[i]); //displaying data in lcd  
    }  
    while(1);  
    return;  
}
```

Let's make use of all these functions and achieve our goal!



```

#include <xc.h>
#define _XTAL_FREQ 6000000
void init();
void lcd_command(unsigned char );
void lcd_data(unsigned char);
unsigned char array[18]="Hello World";
void main(void) {
    init(); //Initialization of LCD
    lcd_command(0x80); // Setting initial location
    for(int i=0;i<11;i++){
        lcd_data(array[i]); //displaying data in lcd
    }
    while(1);
    return;
}

```

```

void init(){
    TRISC = 0x00;//setting all pins as output
    TRISD = 0x00;//setting all pins as output
    delay_ms(100);
    lcd_command(0x30);//Function Set Command for 8 bit
    delay_ms(100);
    lcd_command(0x30);//Function Set Command for 8 bit
    delay_ms(100);
    lcd_command(8);//Function Set Command for 8 bit
    lcd_command(0x38);//Set lines(N) & font(F)
    lcd_command(0x0C);//Display on,cursor off ,blink off
    lcd_command(0x01);//Clear Display
}

```

```

void lcd_command(unsigned char i){
    RC3 = 0;    //RS pin set high (command mode)
    PORTD = i;
    RC0 = 1;    //Enable pin for clock set high
    delay_ms(100);
    RC0 = 0;    //Enable pin for clock set low
}

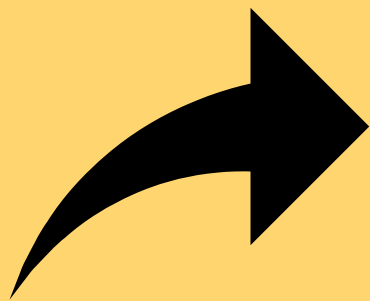
```

```

void lcd_data(unsigned char i ){
    RC3 = 1;    //RS pin set high (data mode)
    PORTD = i;
    RC0 = 1;    //Enable pin for clock set high
    delay_ms(100);
    RC0 = 0;    //Enable pin for clock set low
}

```

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